Fiscal Year:	FY 2015	**Task Last Updated:**	FY 03/23/2015
PI Name:	Oates, Chris Ph.D.		
Project Title:	Development/Evaluation of a 10^-17 Yb Optical Clock for Space Applications		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	chris.oates@nist.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	303-497-7654
Organization Name:	National Institute of Standards and Technology		
PI Address 1:	325 Broadway Street		
PI Address 2:	Mail Stop 847.10		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/01/2012	**End Date:**	05/31/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
Contact Email:	ulf.e.israelsson@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Diddams, Scott (National Institute of Standards & Technology) Ludlow, Andrew (National Institute of Standards & Technology)		
Grant/Contract No.:	NNH12AT81I		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The primary goal of the European Space Agency's (ESA) Space Optical Clock (SOC) program is to develop a high-performance optical frequency standard (clock) for operation in space with 1 x 10^-17 frequency uncertainty. National Institute of Standards and Technology (NIST) proposes to leverage its existing research on world-leading Yb lattice optical clocks to: (1) demonstrate a laboratory-based Yb lattice optical clock with frequency uncertainty of 1 x 10^-17 or better; (2) stabilize a space-friendly fiber frequency comb to the Yb clock signal, and (3) develop a preliminary design for space-qualification by providing clock subsystem specifications. The proposed program will provide ESA with detailed information to make the most informed decisions on use of one or more optical frequency standards for the flight portion of the SOC program. Currently, ESA has identified the Sr and Yb lattice optical clocks as the most promising technologies. A Yb lattice clock has significant potential advantages over Sr in that Yb clocks can operate with solid state lasers much more likely to be practical in space operations. Several European groups are		

presently evaluating Sr's space prospects and Sr has thus received significant attention. For Yb, however, the most mature and high performance lattice clock resides in our lab at NIST. This Yb clock currently performs at least as well as the world's best Sr clocks, approximately 10^-16 frequency uncertainty in all cases, and continues to rapidly improve. Thus, we are uniquely qualified to play a vital role in the SOC project. For a modest investment, the proposed research program will ensure ESA has the best information and options to select the optical clock technology to meet SOC program goals, ensuring fully informed decisions affecting a much larger financial and scientific investment in the SOC program. Furthermore, by demonstrating a 10^-17 level Yb optical lattice clock, our proposed effort will achieve significant milestones currently outlined in the SOC program.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The NIST SOC program (Yb lattice clock) also directly supports the major benefits of the overall Atomic Clock Ensemble in Space (ACES) program by developing a new high performance NIST optical frequency standard that will directly contribute to NIST's roles in the broader ACES program:
1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements. The NIST Yb lattice clock represents a dramatic improvement in stability and accuracy compared to cesium microwave primary frequency standards.

2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory.

3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, search for possible Lorentz transformation variations, and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosomology.

Progress in the most recent reporting period (April 2014 through May 2015) has been substantial, with several tasks ahead of schedule, including the following major advances:
1. Insertion of a blackbody chamber leading to a subsequent reduction of the blackbody uncertainty to less than 1 x 10^-18, fractionally (Yb-1).

2. A series of significant improvements in both the Yb-1 and Yb-2 clocks and systems to dramatically reduce uncertainties and improve performance reliability. Specifically a careful study of lattice light shifts, the largest remaining uncertainty for the Yb clock, was reduced to below 2 x 10^-18, fractionally (Yb-2).

3. Development of a new clock laser system that greatly simplifies the connection to the fiber frequency comb.

4. We provided a detailed listing of required laser systems for the Yb lattice clock to aid in the evaluation of the clock system for space applications.

Progress on each task is described below:

1. Reducing the Uncertainty Due to Blackbody Stark Shifts

At the beginning of this reporting period, the largest remaining systematic shift for the Yb clock frequency resulted from Stark shifts caused by the blackbody radiation that surrounds the trapped atoms, which causes fractional frequency shifts in Yb of 3 x 10-15 for a room temperature apparatus. To reduce the uncertainty for this shift, we constructed a special copper enclosure for the lattice-confined atoms. We installed this thermal shield in our vacuum apparatus, complete with five precision resistance temperature sensors, which, through extensive testing, we found to agree at the < 10 mK level. These sensors enabled us to characterize the temperature of the blackbody environment with an uncertainty below 20 mK (sufficient for < 10^-18 fractional clock uncertainty). A key test of the new blackbody environment was to verify that the Yb lattice clock frequency changed with blackbody temperature according to its known coefficient. (We actually measured this coefficient very precisely in early stages of this program.) The measurement procedure consisted of making frequency comparisons between our two Yb lattice clocks, Yb-1 and Yb-2. We kept the frequency of Yb-2 fixed while heating the blackbody enclosure of Yb-1 (with heaters located on the thermal shield) from 293 K to 380 K. We then recorded the frequency difference between clocks to determine Yb-1's temperature dependence. By synchronizing the measurement cycles of the two, we could determine the frequency difference between the clocks with a fractional uncertainty below 10^-17 in fewer than 10 minutes.

Initial measurements indicated that the temperature sensors yielded inconsistent results at higher temperatures when used in a high vacuum environment, but modified versions of these sensors (that had the gap between the sensor and its container filled with thermal conducting material) resolved this issue. From ten data runs, we derived a set of values for the blackbody coefficient (i.e., the frequency shift vs. temperature), which was in good agreement with the value we derived earlier via dc Stark and other measurements. This gave us a high degree of confidence in our ability to characterize the temperature of the blackbody environment at a level that enabled us to reduce the fractional uncertainty in the Yb clock due to the blackbody correction to below 1 x 10^-18 (one order of magnitude below the requirement for the ESA Space Optical Program goal!). We note that this total blackbody uncertainty is actually principally due to an uncertainty in an atomic structure coefficient – the uncertainty in the temperature of the shield contributes a fractional

Task Progress:	uncertainty of 5.5 x 10^-19. 2. Reducing the Uncertainty of Lattice Light Stark Shifts The final key systematic effect in our effort to report a new (i.e., sub 10^-17) uncertainty budget for the NIST Yb optical lattice clock was the Stark shift associated with the lattice light itself. In a certain sense this shift due to the lattice light is a necessary evil in that it is also responsible for the dipole trapping force that confines the atoms in the lattice. The trick that makes lattice clocks work, of course, is that we can tune the lattice to a particular value, termed the magic wavelength, at which the ground and excited states of the clock transition are equally shifted, thereby leaving the clock transition unperturbed. However, this cancellation is only approximate due to higher-order effects such as the hyper-polarizability (associated with two-photon transitions) and magnetic dipole or electric quadrupole transitions. These effects are small (fractionally below 10^-16), but need to be evaluated and/or minimized for fractional performance below 10^-17. Thus, during this reporting period, we performed a detailed investigation of these contributions. To improve the precision of this evaluation, we leveraged the shifts by means of a build-up cavity, which increased both the intensity of the lattice standing wave (x7) and the trap volume (x10). We then varied the trap depth over a range of lattice frequencies near the magic wavelength and measured the resulting shifts. A critical part of this work was to make sure that changes in the number of atoms trapped in the lattice (that result from changes in trap depth) did not contaminate the measurements. Variations in lattice atom number changed the atom density, which in turn slightly modified the clock frequency. Fortunately, with the build-up cavity, we could use a larger beam waist for the lattice, thereby substantially reducing the atom density, so that the size of the shift was, fractionally, only 6 x 10^-18 (more than an order of magnitude below previous values due to the increased trap volume) with an uncertainty below 10^-18. With the density shifts well understood, we then measured the Stark shifts. We were able to accumulate the data with a single lattice clock system by alternating between two lattice depth values and measuring the frequency differences. At low intensities the linear effect dominated, while the quadratic effect began to take over at lattice depths above 1000 atomic recoils. A series of such plots at different lattice detunings enabled us to find more precisely the magic wavelength and the value of the coefficients for higher-order effects. We found that the second-order shift showed a dependence on trap depth that was correlated with atom temperature, but this temperature sensitivity did not present a significant barrier to 10^-18 performance. After correcting for small shifts due to atom density, motion, and temperature that also change with trap depth (via careful modeling of the lattice trapping environment), we have fit simple second-order polynomials to the data and determined the respective coefficients. Our current level of understanding of the second-order contribution yields a fractional contribution to the clock uncertainty below 10^-18, while the first-order contribution is < 2 x 10^-18. We anticipate further data runs will push this uncertainty below one part in 10^18. 3. New Laser System for Generating Clock Radiation at 578 nm Over the past year we made considerable progress on another important goal of the NIST effort in the NASA Fundamental Physics program, namely the development of a new clock laser system (at half the clock frequency) that can interact much more robustly with the frequency comb. Historically we have generated the 578 nm clock light through sum frequency of 1030 nm and 1319 nm sources. We then stabilized this light through (1) locking to a high finesse cavity at 578 nm and (2) correcting slow cavity drifts by locking to the narrow 578 nm transition in Yb, by feeding back to the 1030 nm laser. However, while this approach has yielded extremely stable 578 nm light, the individual infrared sources themselves are not necessarily stable. As a result, we had to send the stable 578 nm light to the fs-laser frequency combs for measuring, etc. Unfortunately, this wavelength is not well matched to the frequency comb spectra, especially for the infrared comb devices that are most likely to be space qualified. So an important aspect of our NASA program is to demonstrate a laser source that would be more compatible with compact, robust, frequency combs. To this end we developed a clock laser at 1156 nm (half of the clock frequency) based on commercial semi-conductor technology. We note that an additional benefit of working at this frequency is that we can take advantage of the higher reflectivity (and anecdotally, more robust) coatings available in the infrared part of the spectrum. The first step of this clock development project was to pre-stabilize the frequency of this laser to a resonance of a high-finesse 1156 nm reference cavity (length = 29 cm). The cavity has a linewidth of ~ 1 kHz, with a theoretical thermal noise floor of about 7 x 10^-17, which should enable Yb clock stability near 10^-16t^-1/2. We then completed the key step of doubling the frequency of the infrared light to the optical through construction and optimization of a frequency doubling apparatus. The system uses a periodically-poled LiNbO3 crystal in a single-pass configuration. This system doubles the output from a semiconductor laser, and has produced ~ 15 mW output power from only 75 mW of infrared power. With the laser stabilized and the light doubled to the visible domain, we are now poised to begin Yb clock spectroscopy with our new laser source. We also plan to verify the performance of the new 578 nm system via direct comparisons with the existing clock laser.
Bibliography Type:	Description: (Last Updated: 04/05/2023)
Articles in Peer-reviewed Journals	Beloy K, Hinkley N, Phillips NB, Sherman JA, Schioppo M, Lehman J, Feldman A, Hanssen LM, Oates CW, Ludlow AD. "Atomic clock with 1 x 10^-18 room-temperature blackbody Stark uncertainty." Physical Review Letters. 2014 Dec 31;113(26):260801. http://dx.doi.org/10.1103/PhysRevLett.113.260801 , Dec-2014
Articles in Peer-reviewed Journals	Oates CW, Ludlow AD. "Optical Lattice Clocks." Optics and Photonics News. 2015 Jan:36-43. http://www.osa-opn.org/home/articles/volume_26/january_2015/features/optical_lattice_clocks/ ; accessed 3/25/15. , Jan-2015
Articles in Peer-reviewed Journals	Zanon-Willette T, Almonacil S, de Clercq E, Ludlow AD, Arimondo E. "Quantum engineering of atomic phase shifts in optical clocks." Physical Review A. 2014 Nov;90(5):053427. http://dx.doi.org/10.1103/PhysRevA.90.053427 , Nov-2014
Awards	Quinlan F (Franklyn Quinlan). "2015 European Frequency and Time Forum Young Scientist Award, presented at the joint IEEE Frequency Control Symposium / European Frequency and Time Forum meeting in Denver, CO, April 2015." Apr-2015

Awards	Ludlow AD, Oates CW. "2014 Rocky Mountain Eagle Scientific Project Award, given by the Colorado Federal Executive Board, May 2014." May-2014
Awards	Ludlow AD, Oates CW. "2014 Department of Commerce Gold Medal, for 'exceptional scientific creativity and achievement in inventing and perfecting new types of atomic clocks with best in the world performance.' Novembr 2014." Nov-2014